# Python Project – Network and Port Scanner

# Ethics & Detection Report

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## Introduction

For this project, our team built and tested a Python network and port scanner. Because port scanning can be sensitive, we did all our work inside a private lab environment only, as required by the assignment. The goal of this report is to explain where we tested, what port scanning looks like from a defender's point of view, and why authorization matters before using a scanner.

# 1. Testing environment

We only tested our tool in a controlled virtual lab. Our target machine was Metasploitable 2, which is an intentionally vulnerable virtual machine made for security training. We used it as the machine to scan, and we used another machine as the scanner. In our Wireshark observations, the scanning machine was 192.168.163.128 and the Metasploitable target was 192.168.163.130. During host discovery, we also saw another live host at 192.168.163.1 replying to ARP traffic.

This means our testing stayed inside a local, isolated environment and did not target random machines on the internet. That is also what the assignment asked us to do: all testing had to stay inside the lab and not be used on systems we do not own or control.

Our setup was ethically appropriate because:

- we used a training VM designed for this purpose
- we worked inside a lab
- we did not scan external or third-party systems

# 2. Defender's perspective

From a defender's point of view, port scanning is often easy to recognize because it creates a very repetitive pattern on the network.

**2.1 Host discovery**

Before scanning ports, we used host discovery. In Wireshark, this appeared as a large number of ARP requests sent across the local subnet. These requests were broadcast messages asking which device owned a given IP address. We then saw an ARP reply from another machine saying that 192.168.163.1 was active and giving its MAC address.

From the defender's side, this kind of behavior looks suspicious because one machine is not just talking to one known device, it is asking about many devices on the network very quickly.

### 2.2 Port scanning

When we ran the port scan on Metasploitable with Nmap, Wireshark showed the scanner sending TCP SYN packets to many different ports on the same host. The scan started with well-known ports such as 22, 80, 21, 25, 23, and 53, then continued with other ports. This makes the traffic look different from normal user traffic, because a normal user usually connects to one or two services, not dozens of ports one after another.

A defender looking at this traffic would notice:

- many connection attempts from one source
- many destination ports on the same host
- very short connection attempts
- a repeated and systematic pattern

## 3. Importance of authorization

Authorization matters because a scanner is a powerful tool. In a lab, it is a normal learning and security tool. Outside a lab, it can be seen as unauthorized reconnaissance.

The Centre for Cybersecurity Belgium explains that unauthorized access or attempted access to a computer system can fall under Article 550bis of the Criminal Code. The same official guidance also says that even security research must stay necessary and proportionate, and must be done without fraudulent intent or intent to harm if someone wants legal protection.

The CCB also states clearly that unauthorized intrusion into a third-party computer system, even with good intentions, is a criminal offence unless there is authorization or another valid legal framework.

## 5. Compliance

Our project stayed inside the allowed scope for the assignment and inside a controlled educational lab.

- we tested only on Metasploitable 2, a purposely vulnerable training VM
- we worked inside a private local lab
- the assignment clearly required lab-only testing
- we did not scan public IPs or external organizations
- the purpose was educational and defensive, not harmful

We respected both the project instructions and the basic Belgian principle that security testing must not become unauthorized access to third-party systems. Official Belgian guidance links unauthorized access and attempted access to criminal-law provisions, which is exactly why keeping this work inside a lab was important.

## Conclusion

This project showed us that port scanning is not invisible. From a defender's point of view, it creates a recognizable pattern: ARP discovery traffic, repeated SYN packets, and different answers for open and closed ports. In Wireshark, we clearly saw that open ports answered with SYN-ACK, while closed ports answered with RST-ACK.

The project also showed why ethics matters in cybersecurity. A scanner is useful for learning and defense, but it must only be used in an authorized environment. By working only inside our own virtual lab with Metasploitable 2, we kept the project within a safe and lawful scope.